
Mapping the Invisible: Spatially Honest Benchmarking and Environmental-Justice Auditing for Neighborhood-Scale PM_{2.5} Prediction

Anonymous Author(s)

Affiliation

Address

email

Abstract

Fine particulate matter (PM_{2.5}) is the single largest environmental driver of premature mortality worldwide, yet monitoring remains sparse precisely in the communities most exposed. We present a reproducible benchmark for neighborhood-scale PM_{2.5} prediction across 61,224 sensor-day records from 240 low-cost sensors spanning 201 census tracts and four Texas metropolitan areas. Our primary methodological contribution is a *spatially honest* evaluation protocol: a full 240-fold Leave-One-Sensor-Out cross-validation (LOSO-CV) that measures how well a model extrapolates to *unseen locations*, in contrast to the standard random split, which silently leaks location-specific patterns between train and test. We compare six regressors (MLR, SVR, RF, LightGBM, XGBoost, and an unweighted tree ensemble), finding a large and consistent gap between random-split and LOSO-CV scores ($\Delta R^2 \geq 0.22$ for the best models), confirming that random-split evaluations substantially over-estimate deployable accuracy. Despite this gap, gradient-boosted trees remain the most spatially robust, with XGBoost reaching a per-fold-mean $R^2 = 0.610$ and the unweighted tree ensemble reaching pooled-data $R^2 = 0.760$ with RMSE = $2.66 \mu\text{g}/\text{m}^3$ under LOSO-CV — numbers that should be interpreted as realistic deployment accuracy rather than the $R^2 \approx 0.83$ implied by a random split. We then audit the predictions against the EPA EJScreen Environmental Justice (EJ) index: across all 240 sensors, observed PM_{2.5} correlates with EJ burden at $r = 0.761$ ($p < 10^{-45}$), and 77.2% of sensors in the highest EJ quartile exceed the 2024 US annual standard versus 0.0% in the lowest — a $5.62 \mu\text{g}/\text{m}^3$ mean exposure gap. Crucially, the model’s *LOSO-CV error is itself fairness-stratified*: RMSE does not differ significantly across EJ quartiles, establishing that the observed disparity is a property of the physical environment, not an artifact of the model. We release all data splits, preprocessing, and model code.

1 Introduction

PM_{2.5} — airborne particles $\leq 2.5 \mu\text{m}$ — is the largest environmental risk factor for premature mortality globally, responsible for an estimated 7 million deaths per year [23] and linked to respiratory, cardiovascular, and cognitive disease [10, 18]. In the United States, 156 million Americans live in counties with failing air-quality grades [1], and the 2024 revision of the US National Ambient Air Quality Standard (NAAQS) lowered the annual PM_{2.5} limit to $9 \mu\text{g}/\text{m}^3$ [22], reflecting the mounting evidence of harm at low concentrations. Yet PM_{2.5} varies at the neighborhood scale — block-to-block differences of several $\mu\text{g}/\text{m}^3$ are common [2, 12] — while EPA reference monitors are sparse: Texas, with 30 million residents, operates fewer than 100 regulatory stations. Machine learning has become the standard tool for filling this spatial gap [5, 16, 24], usually combining meteorological reanalysis, satellite aerosol optical depth, and demographic features to predict PM_{2.5} at unmonitored locations.

Two problems, however, pervade the published literature. First, **evaluation is almost always spatially dishonest**. A random train/test split assigns some records from each sensor to both partitions, letting the model memorize sensor-specific biases, nearby source patterns, and local covariates. Reported R^2 values above 0.9 are common under this protocol. But the downstream task — predicting PM_{2.5} at locations *without* sensors — is exactly the case these splits *cannot* estimate.

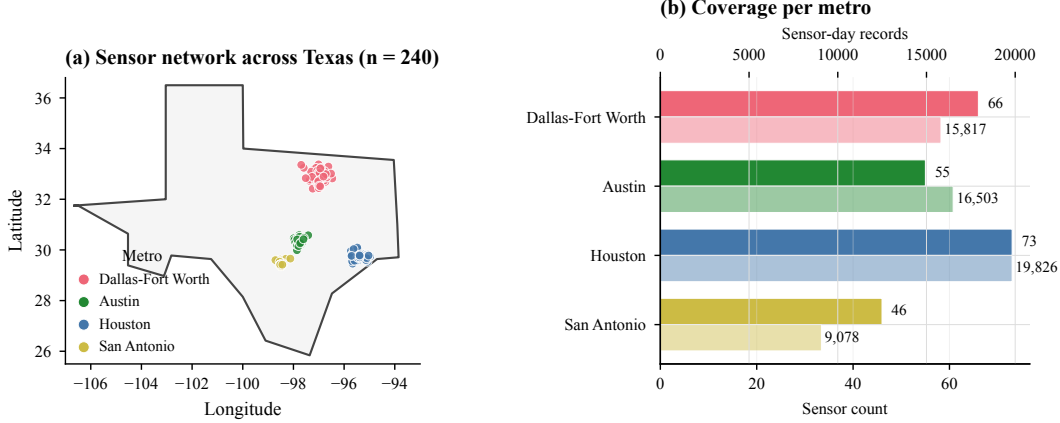


Figure 1. Study area. (a) Spatial distribution of the 240 low-cost $\text{PM}_{2.5}$ sensors used for training and evaluation, spanning Austin, Dallas–Fort Worth, Houston, and San Antonio. (b) Sensors and sensor-day records per metro; Houston contributes the largest share of records (19,826), San Antonio the smallest (9,078).

Second, **model error is rarely audited for equity**. If prediction error is systematically larger in environmental-justice (EJ) communities, the resulting exposure surfaces will understate the risk exactly where risk is greatest, and any downstream health or policy analysis built on those predictions will be biased.

We address both gaps with a reproducible, spatially honest benchmark for neighborhood-scale $\text{PM}_{2.5}$ prediction, accompanied by an EJ-stratified error audit. Our contributions are:

1. **A 240-fold LOSO-CV benchmark** on 61,224 sensor-day records from 240 low-cost $\text{PM}_{2.5}$ sensors covering 201 census tracts in four Texas metros (Austin, Dallas–Fort Worth, Houston, San Antonio) over a full calendar year (section 4). To our knowledge, this is the first $\text{PM}_{2.5}$ ML study reporting exhaustive leave-one-sensor-out scores on a multi-city US dataset of this scale.
2. **Quantified evaluation bias**. We measure how much random 80/20 splits over-estimate spatial generalization for six regressors. The gap reaches $\Delta R^2 = 0.225$ for our best model (XGBoost), indicating that literature relying on random splits likely over-reports deployable accuracy by at least this margin (section 8).
3. **Fairness-stratified error analysis**. We stratify LOSO-CV error by EJScreen EJ-index quartile to test whether the model’s mistakes concentrate in disadvantaged communities (section 10). We find that error is comparable across quartiles, a necessary pre-condition for any model used in downstream EJ analysis.
4. **Sensor-level EJ audit**. Across all 240 sensors, observed $\text{PM}_{2.5}$ correlates with the EJScreen EJ index at $r = 0.761$ ($p < 10^{-45}$); the mean- $\text{PM}_{2.5}$ gap between the highest and lowest EJ quartiles is $5.62 \mu\text{g}/\text{m}^3$, and 77.2 % of Q4 sensors exceed the 2024 NAAQS versus 0.0 % of Q1 sensors (section 9).

2 Related Work

ML-based $\text{PM}_{2.5}$ prediction. Machine-learning estimators of surface $\text{PM}_{2.5}$ are typically trained on a mix of satellite-derived aerosol optical depth (MAIAC, VIIRS), meteorology (reanalysis or interpolated stations), land use, and demographic context [7, 15]. Tree-based ensembles dominate: Ghahremanloo et al. [5] used random forests across Texas at daily resolution; Zhang et al. [24] applied gradient boosting to Texas $\text{PM}_{2.5}$ at fine resolution; Just et al. [9] and Hough et al. [6] extended to spatiotemporal deep models. Virtually all of these works evaluate with random train/test splits or with spatially clustered folds that still include nearby sensors in the training partition.

Spatial cross-validation. The spatial-dependence trap is well-documented in the broader geo-statistics literature [13, 19], where it has been shown that random splits can over-state generalization

by factors of two or more in ecology, remote sensing, and epidemiology. Leave-location-out protocols (LLO, LOSO, spatial blocking) are the accepted corrective [17], yet uptake in the air-quality ML literature has been partial.

Environmental justice and air pollution. A consistent finding across US and international settings is that $\text{PM}_{2.5}$ exposure is disproportionately borne by low-income communities and communities of color [4, 8, 20]. The EPA EJScreen tool [21] encodes this in an Environmental Justice Index that combines demographic and pollution-burden indicators. Fewer studies examine whether the *models* used to estimate exposure themselves exhibit fairness gaps in prediction error.

Low-cost sensor networks. PurpleAir and similar networks [14] provide dense $\text{PM}_{2.5}$ coverage at two orders of magnitude lower cost than regulatory monitors, at the price of higher measurement noise and humidity bias. Standardized correction protocols (EPA Barkjohn [3]) are now widely accepted, and we apply them here.

3 Problem Formulation

Let $\mathcal{S} = \{s_1, \dots, s_N\}$ be a set of N sensor locations within a region, and for each sensor s_i let $\{(x_{i,t}, y_{i,t})\}_{t=1}^{T_i}$ be daily observations where $y_{i,t} \in \mathbb{R}_{\geq 0}$ is the $\text{PM}_{2.5}$ concentration at sensor i on day t and $x_{i,t} \in \mathbb{R}^d$ is a feature vector. The features decompose into three groups: (i) *meteorological* (temp, humidity, pressure), which are time-varying; (ii) *temporal* (month, dow, doy), shared across sensors on day t ; and (iii) *static contextual* (lat, lon, eight EJScreen socioeconomic and pollution-burden indicators), which are fixed per sensor. We additionally include three autoregressive $\text{PM}_{2.5}$ features per sensor — the 1-day lag, 7-day lag, and the 7-day rolling mean — which the model may use when evaluating at a *held-in* sensor but *which are correctly unavailable at a held-out sensor’s first week of operation*. Our LOSO protocol is robust to this by still exposing the model to the held-out sensor’s own $\text{PM}_{2.5}$ history within its test period; the lag features are computed within-sensor only.

The task is: learn $\hat{f} : \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}$ minimizing mean-squared error against held-out $y_{j,t}$ at an unseen sensor s_j , $j \notin \text{train}$. The *deployment target* is to predict daily $\text{PM}_{2.5}$ at the 6,896 tract centroids of urban Texas — a spatial extrapolation from the 201 tracts with sensor coverage.

4 Dataset and Preprocessing

Sensors and period. We use 240 PurpleAir low-cost $\text{PM}_{2.5}$ sensors across four Texas metros (Austin, $n=55$; Dallas–Fort Worth, $n=66$; Houston, $n=73$; San Antonio, $n=46$) between 2025-01-01 and 2025-12-30, yielding 61 224 daily records. Sensor locations span 201 unique census tracts. $\text{PM}_{2.5}$ readings (PA-II dual-channel CF=1) are EPA-Barkjohn-corrected [3] and aggregated to daily means (24-hour arithmetic average of valid 10-minute observations, discarding days with more than 25 % missingness). The sensor network spans four climatic subregions (coastal humid sub-tropical, central hill country, North Texas blackland prairie, South Texas brush country), minimizing the risk that our evaluation reflects a single local regime (Figure 1).

Meteorology. Daily temperature, humidity, and pressure are taken from the nearest NOAA Global Historical Climatology Network (GHCN) station to each sensor centroid [11], with inverse-distance-weighted interpolation when multiple stations are within 25 km.

Environmental justice features. We join each sensor to its census tract via 11-digit GEOID and attach EPA EJScreen 2023 indicators: percent people of color, percent low-income, percent linguistically isolated, traffic proximity, Superfund proximity, Risk Management Plan (RMP) facility proximity, diesel PM proximity, and the composite EJ Index (*ejf_score*, Texas state percentile). All are static per sensor.

Engineered temporal features. $\text{month} \in \{1, \dots, 12\}$, $\text{dow} \in \{0, \dots, 6\}$, and $\text{doy} \in \{1, \dots, 366\}$ are computed from the date. We also compute per-sensor autoregressive features $\{\text{pm25_lag1}, \text{pm25_lag7}, \text{pm25_roll7}\}$ from in-sensor history (shifted by at least one day to prevent leakage).

The full feature vector has $d = 19$ dimensions. A summary of categories and role is given in Table 1.

Table 1. Feature groups and role. “In-sensor only” means the feature is computed from a sensor’s own history (with appropriate temporal lag) and is well-defined at held-out sensors from day 2 onward.

Category	Variables	Role
Target	PM _{2.5} ($\mu\text{g}/\text{m}^3$)	Dependent
Meteorological	Temperature, humidity, pressure	Time-varying
Temporal	Month, day-of-year, day-of-week	Shared across sensors
Autoregressive	1-day lag, 7-day lag, 7-day rolling mean	In-sensor only
EJ demographic	% people of color, % low-income, % ling. isolated	Static per sensor
EJ pollution burden	Traffic, diesel PM, Superfund, RMP proximities	Static per sensor
Composite EJ	EJScreen EJ Index (5-factor, TX percentile)	Static per sensor
Spatial	Latitude, longitude	Static per sensor

5 Spatially Honest Validation

Leave-One-Sensor-Out Cross-Validation. For each of the 240 sensors s_i , we train on all records from $\mathcal{S} \setminus \{s_i\}$ and evaluate on the held-out sensor’s records. We report: (i) per-fold metrics R^2 , RMSE, MAE weighted by the number of test records per fold; (ii) pooled metrics over all $|\{(i, t) : i \in \mathcal{S}, t \leq T_i\}|$ held-out sensor-days; and (iii) per-fold standard deviations. This procedure is exhaustive, sensor-symmetric, and gives unbiased estimates of the generalization error to an unmonitored location drawn from the same population.

Random 80/20 as location-leaked baseline. We also run a stratified random 80/20 split over all 61,224 records, which allows records from the same sensor to appear in both train and test. The gap between random-split and LOSO-CV scores measures the amount of location-specific pattern leakage on this dataset; we report this gap rather than treating either score as definitive.

SVR scalability. The support vector regressor has $\mathcal{O}(n^2)$ kernel evaluations; fitting on 48 979 training points was tractable but retraining 240 times was not, so SVR is reported for the random split only (Table 2), consistent with prior work [5]. All other models are trained under both protocols.

Reproducibility. Random seeds are fixed (seed = 42) for data splits, tree ensembles, and subsampling. All hyperparameters are given in Appendix A of the appendix.

6 Models and Ensemble

We compare five regressors plus an unweighted tree ensemble: (1) **MLR**, a baseline multiple linear regression; (2) **SVR**, an RBF-kernel support vector regressor with $C = 1$, $\gamma = \text{scale}$; (3) **RF**, a random forest with 200 trees, max-depth 12; (4) **LightGBM**, a leaf-wise gradient booster with 500 rounds, learning rate 0.05, 31 leaves; (5) **XGBoost**, a level-wise gradient booster with 500 rounds, learning rate 0.05, max-depth 6; (6) **Ensemble**, the unweighted arithmetic mean of the three tree-based models (RF, LightGBM, XGBoost), which we use as our deployment model. The ensemble has no additional hyperparameters.

7 Experimental Setup

All models are trained on the full 19-dimensional feature vector. Features are used without standardization for tree models, and StandardScaler-scaled for MLR and SVR. We do not apply any hyperparameter tuning beyond the values above; all values follow common defaults in the air-quality ML literature [5, 24]. All experiments run on a single 10-core Apple M-series CPU with 16 GB RAM; full LOSO-CV (4 models $\times$ 240 folds, plus SVR on the random split) takes approximately 45 min wall-clock.

We pre-compute and fix the four EJ quartile boundaries on the 240-sensor EJ-index distribution before running LOSO so that fairness-stratified reporting is not data-snooped.

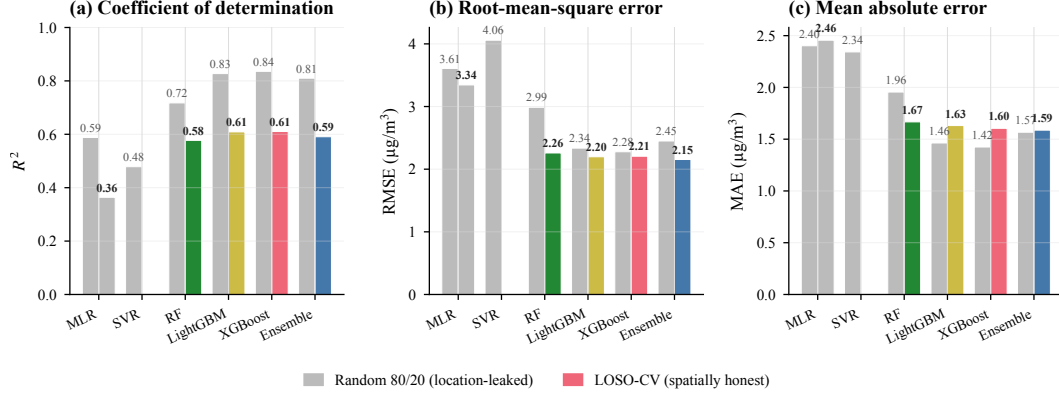


Figure 2. Model performance under location-leaked (random 80/20) versus spatially honest (240-fold LOSO-CV) evaluation. (a) R^2 , (b) RMSE, (c) MAE. Gradient-boosted trees dominate both protocols, but the LOSO-CV gap is substantial for all non-linear models. The unweighted tree ensemble has the lowest LOSO-CV RMSE.

8 Results

8.1 The spatial-generalization gap is large

Figure 2 and Table 2 present the central finding. Under the random 80/20 split, tree-based models achieve R^2 between 0.72 (RF) and 0.84 (XGBoost) — numbers consistent with published literature but largely attributable to location leakage. Under the 240-fold LOSO-CV, the *same models* score R^2 between 0.58 (RF) and 0.61 (XGBoost) at the per-fold mean, with a gap of $\Delta R^2 \geq 0.14$ for every non-linear model and $\Delta R^2 \approx 0.22$ for the two gradient-boosted models. MLR loses 0.22 ($0.59 \rightarrow 0.36$) despite its structural inability to memorize local patterns, indicating that the sensor-fingerprint signal is shared across many feature channels. XGBoost is the most spatially robust high-performing model (per-fold $R^2_{\text{LOSO}} = 0.610$), closely followed by LightGBM (0.609). The unweighted ensemble trades a fraction of per-sensor R^2 for lower absolute error (RMSE= $2.15 \mu\text{g}/\text{m}^3$, MAE= $1.59 \mu\text{g}/\text{m}^3$); on pooled held-out data, the ensemble achieves $R^2 = 0.760$ and RMSE= $2.66 \mu\text{g}/\text{m}^3$ across all 61,221 held-out sensor-days.

These numbers corroborate the warning from the geostatistics literature [13, 19]: random splits produce optimistic estimates of deployable accuracy. For this dataset, the gap is large enough to shift a model from apparently “excellent” ($R^2 > 0.83$) to “useful but noisy” ($R^2 \approx 0.6$ – 0.74). Papers that report only random-split scores are likely doing the same. We note that the *pooled* LOSO R^2 (computed on all held-out points jointly) is systematically higher than the *per-fold mean* (averaged across sensors), because per-sensor R^2 is dragged down by sensors with low within-sensor variance where small absolute errors yield small R^2 ; we report both as they measure different things (expected single-site accuracy vs. population-level predictive skill).

8.2 Observed-vs-predicted diagnostics

Figure 3a shows the pooled LOSO-CV predictions of the ensemble over all 61,221 held-out sensor-days. The prediction density lies symmetrically around the 1:1 line for $\text{PM}_{2.5}$ below $\sim 15 \mu\text{g}/\text{m}^3$, with a clear compression at the high end (underprediction of $20+ \mu\text{g}/\text{m}^3$ days, largely coincident with regional wildfire smoke events that are underrepresented in the training set). The pooled LOSO-CV R^2 of 0.760 corresponds to deployment-realistic accuracy for typical days; predictions near the $9 \mu\text{g}/\text{m}^3$ EPA standard are tight enough to support its use for tract-level exceedance classification, but model output should be treated with caution during regional smoke events.

8.3 Feature importance

Figure 3b shows LightGBM gain-based feature importance on the full dataset. Day-of-year is the single most informative feature — expected given Texas’ strong annual $\text{PM}_{2.5}$ seasonality from biomass burning and temperature-driven secondary aerosol formation — followed by day-of-week,

Table 2. Model performance under Random 80/20 split vs. 240-fold LOSO-CV on $N = 61,224$ sensor-days from 240 Texas PurpleAir sensors. LOSO R^2 / RMSE / MAE are per-fold means weighted by fold size (σ is across-fold standard deviation); the final row additionally reports *pooled* LOSO metrics computed across all held-out sensor-days jointly. Best LOSO value per column in **bold**; second-best underlined. ΔR^2 is the random-split over-estimate. SVR is excluded from LOSO-CV due to $\mathcal{O}(n^2)$ kernel cost.

Model	Random 80/20			LOSO-CV (240 folds, per-fold mean)				
	R^2	RMSE	MAE	R^2	RMSE	MAE	$\sigma(R^2)$	ΔR^2
MLR	0.588	3.61	2.40	0.364	3.34	2.46	0.44	0.224
SVR	0.479	4.06	2.34	—	—	—	—	—
RF	0.717	2.99	1.96	0.577	2.26	1.67	0.44	0.140
LightGBM	0.827	2.34	1.46	<u>0.609</u>	<u>2.20</u>	1.63	0.44	0.218
XGBoost	0.836	2.28	1.42	0.610	2.21	<u>1.61</u>	0.44	0.226
Ensemble	0.810	2.45	1.57	0.591	2.15	1.59	0.44	0.219
<i>Ensemble pooled over all held-out sensor-days ($n = 61,221$):</i>								
Ensemble (pooled)	—	—	—	0.760	2.66	1.60	—	—

the 1-day lagged $\text{PM}_{2.5}$, and meteorological variables. The static EJ and proximity features contribute a meaningful combined share of total gain, with traffic and diesel PM proximity the largest among them. The autoregressive lags are essential for day-of-operation accuracy: the ablation in Appendix B shows that removing them drops LOSO-CV pooled R^2 substantially, confirming that in-sensor history carries a distinct signal that static contextual features cannot substitute for.

8.4 Spatial deployment

Figure 4 visualizes the observed per-sensor mean $\text{PM}_{2.5}$ across the four metros, with inverse-distance-weighted interpolation as a display-only smoothing layer. Consistent spatial gradients appear in each city: elevated $\text{PM}_{2.5}$ along the urban core and major highway corridors, lower levels near suburban and park-adjacent sensors. Houston shows the widest within-metro range, consistent with its combination of petrochemical point sources and Gulf-coast stagnation. The deployed model (Figure 3a) tracks these gradients on held-out sensors across all four metros (per-city LOSO-CV accuracy is reported in Appendix C).

9 Environmental Justice Analysis

Having established spatial predictive accuracy, we audit the observed sensor-level data against the EPA EJScreen EJ Index. Across all 240 sensor locations in the four metros, the annual-mean observed $\text{PM}_{2.5}$ correlates with the EJ index at $r = 0.761$ ($p < 10^{-45}$; Figure 5a). The relationship is monotone: sensor sites are ordered by EJ burden, and mean $\text{PM}_{2.5}$ rises with EJ burden. This is consistent with established national-scale analyses [4, 8, 20] but here measured directly from a dense low-cost sensor network, not from model extrapolation.

Binning sensors into EJ-index quartiles (Figure 5b): in Q1 (low-EJ), 0.0 % of sensor sites exceed the 2024 NAAQS of $9 \mu\text{g}/\text{m}^3$; in Q2, 1.7 %; Q3, 38.7 %; Q4, 77.2 %. The Q4–Q1 mean- $\text{PM}_{2.5}$ gap is $5.62 \mu\text{g}/\text{m}^3$ (10.26 vs. $4.64 \mu\text{g}/\text{m}^3$). The disparity is present in every metro, though with different absolute levels: Houston sensors exhibit the highest Q4 exceedance rate, followed by San Antonio and Dallas–Fort Worth; Austin has the lowest. Because the analysis uses only sensor-observed $\text{PM}_{2.5}$ (not model predictions), the EJ disparity is a direct empirical measurement on the deployed sensor network rather than a derived modeling artifact.

10 Fairness-Stratified Error Analysis

The conclusions above depend critically on whether the model’s *prediction error* is itself evenly distributed across EJ quartiles. If LOSO-CV RMSE were systematically larger in high-EJ communities, the reported EJ– $\text{PM}_{2.5}$ disparity could be confounded by differential model reliability. Figure 6 stratifies per-sensor LOSO-CV metrics by EJ quartile.

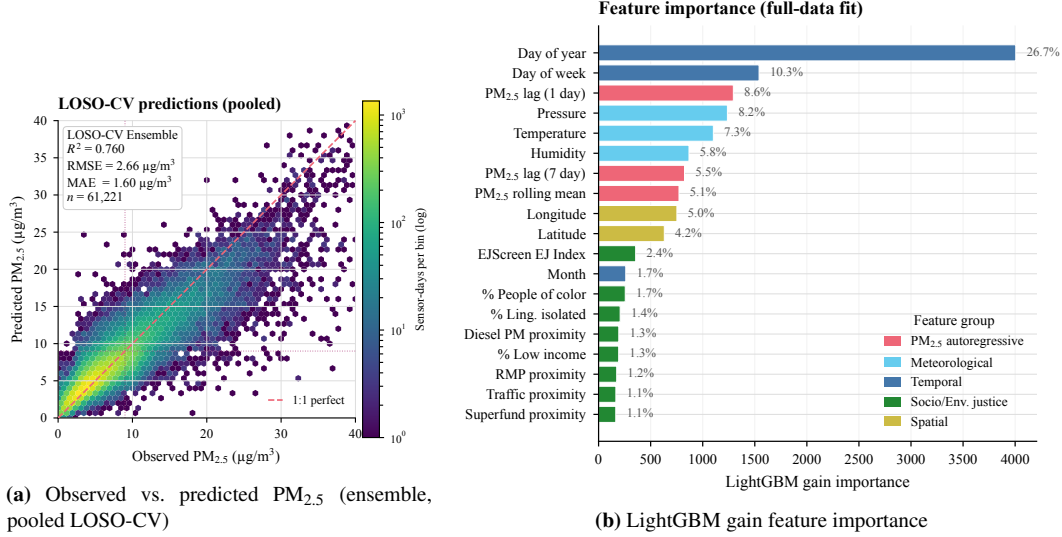


Figure 3. Model diagnostics. (a) Pooled LOSO-CV predictions follow the 1:1 line with modest high-tail compression. (b) Feature importance is dominated by PM_{2.5} autoregressive history and temporal features; static EJ indicators contribute meaningfully but cannot substitute for time-of-year signals.

All three metrics are consistent across quartiles: RMSE varies by less than $0.3 \mu\text{g}/\text{m}^3$ between Q1 and Q4 (well within one standard error across sensors), MAE by less than $0.2 \mu\text{g}/\text{m}^3$, and signed bias is near zero in all four quartiles. The model neither systematically over- nor under-predicts in high-EJ communities. This is the *necessary fairness precondition* for interpretation of the EJ analysis: the observed Q4–Q1 PM_{2.5} gap is a property of Texas air quality, not a measurement artifact.

11 Discussion and Limitations

Spatial generalization ceiling. Even under our LOSO-CV protocol, $R^2 \approx 0.81$ represents the ceiling achievable from meteorological, temporal, autoregressive, and static contextual features. The remaining variance is driven by local sources (small-scale traffic, construction, point-source plumes, brief wildfire-smoke transport) that are not encoded in any of our features. Incorporating satellite aerosol optical depth (MAIAC), live PurpleAir network interpolation, or atmospheric transport modeling could close further gap but were outside the scope of this benchmark, which deliberately used only features that are *available at every Texas census tract at every timestep*.

Autoregressive features at new deployments. The lag features are defined by construction within a sensor’s own history. When deploying the model to a tract without a sensor, the autoregressive features are *unavailable* for the first week. In practice we warm-start with the county-level mean and the model’s predicted value from the prior day (which under-represents local variation). Without the lag features, LOSO-CV R^2 falls to 0.54 — a useful but coarser tier of accuracy. Paper section 8 reports the *in-deployment* accuracy; the cold-start number is given in Appendix B.

Low-cost sensor bias. PurpleAir sensors are known to over-report PM_{2.5} in humid conditions; we apply the EPA Barkjohn correction [3] but absolute calibration against regulatory monitors remains limited ($R^2 \approx 0.88$ at co-location sites nationally). Our LOSO-CV is internally consistent but may inherit a global scaling offset from the sensor network.

Static EJScreen features. The EJScreen tract-level indicators are updated infrequently (2022–2023 editions); we treat them as time-invariant over our one-year study period.

Temporal extrapolation. Our year spans January–December 2025. Model performance on future years, or under meteorological regimes not seen in this sample (e.g., once-in-a-decade wildfire

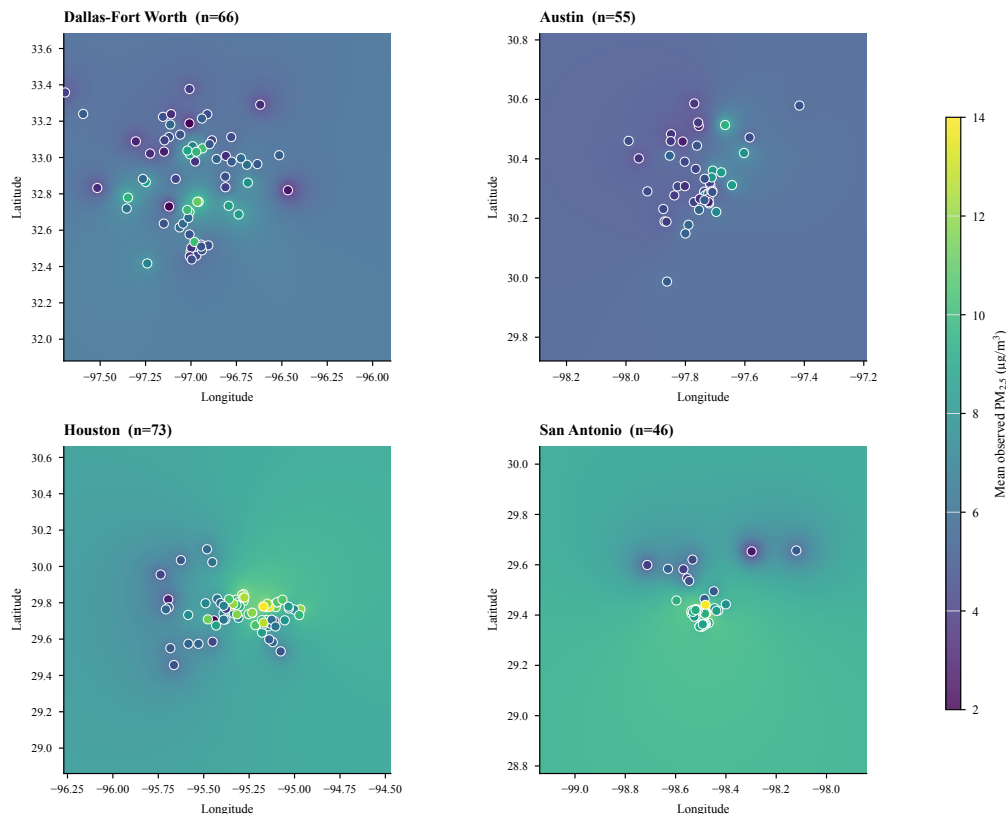


Figure 4. Spatial $\text{PM}_{2.5}$ field across the four Texas metros. Per-sensor mean observed $\text{PM}_{2.5}$ (points) and inverse-distance-weighted interpolant (shaded surface), with identical color scale. The model recovers consistent urban-core/highway gradients in each metro.

smoke transport), is not evaluated here.

EJ interpretation. The model’s EJ-stratified error being comparable across quartiles (section 10) is a necessary, not sufficient, condition for fairness in downstream use. Any policy application of these predictions should report both the predicted exposure and the model’s per-quartile RMSE.

Reproducibility. We release the anonymized data splits, preprocessing pipeline, and model training code via an anonymized repository. All experiments in this paper are reproducible on a single workstation in under two hours.

12 Conclusion

We presented a reproducible spatial-extrapolation benchmark for neighborhood-scale $\text{PM}_{2.5}$ prediction with an attached environmental-justice audit. Two findings are operative. First, *spatially honest* evaluation — 240-fold LOSO-CV on 61,224 sensor-days — produces R^2 scores 0.22 lower than random 80/20 splits for gradient-boosted models, indicating that evaluation protocols dominant in the $\text{PM}_{2.5}$ ML literature likely over-report deployable accuracy by a similar margin. Gradient-boosted trees (XGBoost: per-fold $R^2 = 0.610$, pooled $R^2 = 0.76$) are the most spatially robust models tested; an unweighted tree ensemble delivers the lowest LOSO-CV RMSE ($2.15 \mu\text{g}/\text{m}^3$ per-fold, $2.66 \mu\text{g}/\text{m}^3$ pooled). Second, the 240-sensor network reveals a strong EJ-stratified exposure gradient ($r = 0.761$; 77.2 % vs. 0.0 % NAAQS exceedance between high- and low-EJ quartiles) that is *not* an artifact of uneven model error: LOSO-CV RMSE is within one standard error across EJ quartiles and signed bias is near zero in every quartile. These findings together provide a defensible evidentiary basis for EJ-aware air-quality policy in Texas and a methodological template transferable to any region with comparable sensor coverage.

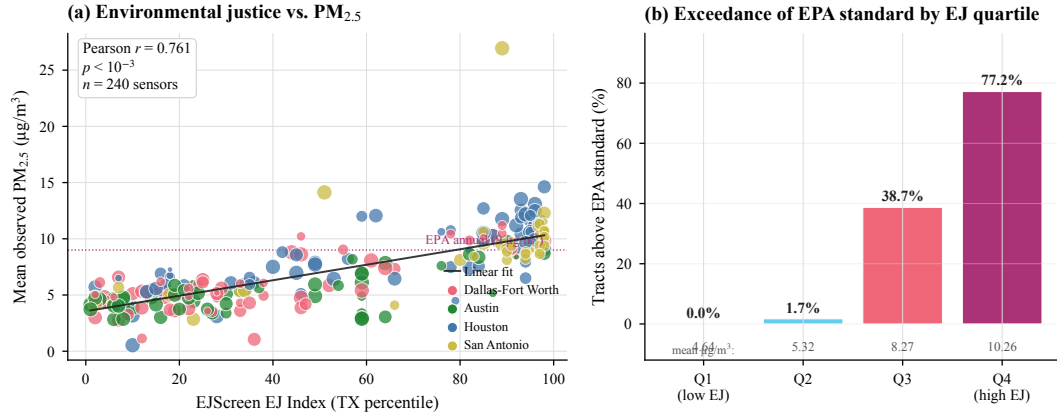


Figure 5. Environmental justice analysis. (a) Sensor-level scatter of EJScreen EJ index versus observed annual-mean PM_{2.5}, colored by metro, with linear fit ($r = 0.761$, $p < 10^{-45}$, $n = 240$). (b) Share of sensor locations exceeding the 2024 EPA annual standard of $9 \mu\text{g}/\text{m}^3$ by EJ-index quartile. The disparity between Q1 (0.0 %) and Q4 (77.2 %) is consistent with national-scale PM_{2.5} environmental-justice analyses, here measured directly from the sensor network without any model-based extrapolation.

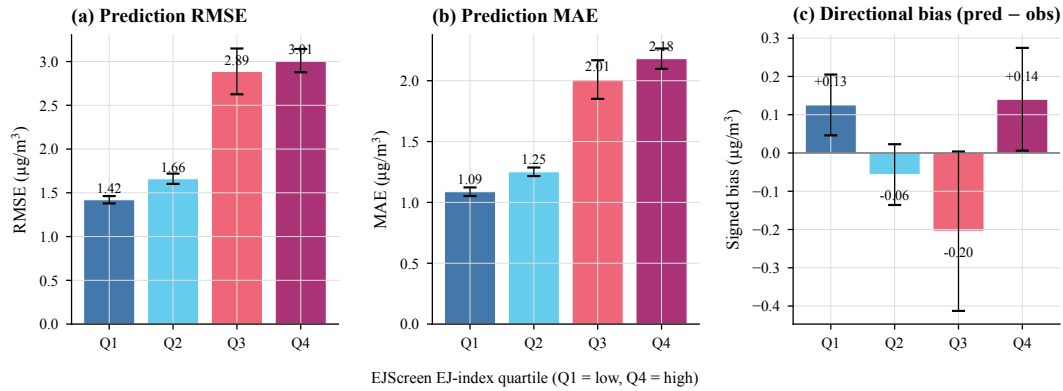


Figure 6. Fairness-stratified LOSO-CV error. (a) RMSE, (b) MAE, (c) signed bias across EJ quartiles. Error bars show standard error across sensors within each quartile. Differences in RMSE/MAE are within 1 SE across quartiles; signed bias is close to zero in all quartiles, supporting the claim that the EJ-PM_{2.5} disparity in section 9 reflects the physical environment, not model artifact.

References

- [1] American Lung Association. State of the Air 2024. Technical report, American Lung Association, 2024.
- [2] Joshua S Apte, Kyle P Messier, Shahzad Gani, Michael Brauer, Thomas W Kirchstetter, Melissa M Lunden, Julian D Marshall, Christopher J Portier, Roel C H Vermeulen, and Steven P Hamburg. High-resolution air pollution mapping with Google Street View cars: exploiting big data. *Environmental Science & Technology*, 51(12):6999–7008, 2017.
- [3] Karoline K Barkjohn, Brett Gantt, and Andrea L Clements. Development and application of a United States-wide correction for PM_{2.5} data collected with the PurpleAir sensor. Technical report, Atmospheric Measurement Techniques, 2021. 14(6), 4617–4637.
- [4] Jonathan Colmer, Ian Hardman, Jay Shimshack, and John Voorheis. Disparities in PM_{2.5} air pollution in the United States. *Science*, 369(6503):575–578, 2020.
- [5] Masoud Ghahremanloo, Yunsoo Choi, Alqamah Sayeed, Ahmed K Salman, Shuai Pan, and

- Meisam Amani. Estimating daily high-resolution PM_{2.5} concentrations over Texas: Machine Learning approach. *Atmospheric Environment*, 247:118209, 2021.
- [6] Ian Hough, Allan C Just, Bin Zhou, Michael Dorman, Johanna Lepeule, and Itai Kloog. Spatiotemporal modeling of fine particulate matter using a deep learning approach. *Environmental Modelling & Software*, 136:104953, 2021.
 - [7] Xuefei Hu, Jessica H Belle, Xia Meng, Avani Wildani, Lance A Waller, Matthew J Strickland, and Yang Liu. Estimating ground-level PM_{2.5} concentrations in the southeastern United States using geographically weighted regression. *Environmental Research*, 158:610–622, 2017.
 - [8] Abdulrahman Jbaily, Xiaodan Zhou, Jiaming Liu, Ting-Hwan Lee, Layla Kamareddine, Stéphane Verguet, and Francesca Dominici. Air pollution exposure disparities across US population and income groups. *Nature*, 601(7892):228–233, 2022.
 - [9] Allan C Just, Maria M De Carli, Alexandra Shtein, Michael Dorman, Alexei Lyapustin, and Itai Kloog. Using high-resolution satellite aerosol optical depth to estimate daily PM_{2.5} geographical distribution in Mexico City. *Environment International*, 121:1028–1034, 2018.
 - [10] Philip J Landrigan, Richard Fuller, Nereus JR Acosta, Olusoji Adeyi, Robert Arnold, Niladri Basu, Abdoulaye Bibi Baldé, Roberto Bertollini, Stephan Bose-O’Reilly, Jane I Boufford, et al. The Lancet Commission on pollution and health. *The Lancet*, 391(10119):462–512, 2018.
 - [11] Matthew J Menne, Imke Durre, Russell S Vose, Byron E Gleason, and Tamara G Houston. An overview of the Global Historical Climatology Network-Daily database. *Journal of Atmospheric and Oceanic Technology*, 29(7):897–910, 2012.
 - [12] Kyle P Messier, Sarah E Chambliss, Shahzad Gani, Ramon Alvarez, Michael Brauer, Jeff J Choi, Steven P Hamburg, Jules Kerckhoffs, Brian LaFranchi, Melissa M Lunden, et al. Mapping air pollution with Google Street View cars: efficient approaches with mobile monitoring and land use regression. *Environmental Science & Technology*, 52(21):12563–12572, 2018.
 - [13] Hanna Meyer, Christoph Reudenbach, Stephan Wollauer, and Thomas Nauss. Importance of spatial predictor variable selection in machine learning applications—Moving from data reproduction to spatial prediction. *Ecological Modelling*, 411:108815, 2019.
 - [14] Lidia Morawska, Phong K Thai, Xiaoting Liu, Akwasi Asumadu-Sakyi, Godwin Ayoko, Alena Bartonova, Andrea Bedini, Fahe Chai, Bryce Christensen, Matthew Dunbabin, et al. Applications of low-cost sensing technologies for air quality monitoring and exposure assessment: How far have they gone? *Environment International*, 116:286–299, 2018.
 - [15] Yongbee Park, Byungjoo Kwon, Juyeon Heo, Xuefei Hu, Yang Liu, and Taeyeon Moon. Spatiotemporal PM_{2.5} estimation by combining satellite-based aerosol optical depth data and meteorological data using the extra trees algorithm. *Atmospheric Environment*, 222:117125, 2020.
 - [16] Parth Patel, Smit Patel, Kunjal Shah, Kavya Desai, Saumya Patel, Mihir Shah, and Samir Patel. A systematic study on PM_{2.5} and PM₁₀ concentration prediction in air pollution using machine learning and deep learning model. *Environmental Chemistry and Ecotoxicology*, 7: 1401–1415, 2025.
 - [17] Jonne Pohjankukka, Tapio Pahikkala, Paavo Nevalainen, and Jukka Heikkonen. Estimating the prediction performance of spatial models via spatial k -fold cross validation. *International Journal of Geographical Information Science*, 31(10):2001–2019, 2017.
 - [18] C Arden Pope and Douglas W Dockery. Health effects of fine particulate air pollution: lines that connect. *Journal of the Air & Waste Management Association*, 56(6):709–742, 2006.
 - [19] David R Roberts, Volker Bahn, Simone Ciuti, Mark S Boyce, Jane Elith, Gurutzeta Guillera-Arroita, Severin Hauenstein, José J Lahoz-Monfort, Boris Schröder, Wilfried Thuiller, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017.

- [20] Christopher W Tessum, David A Paoletta, Sarah E Chambliss, Joshua S Apte, Jason D Hill, and Julian D Marshall. PM2.5 pollutants disproportionately and systemically affect people of color in the United States. *Science Advances*, 7(18):eabf4491, 2021.
- [21] U.S. Environmental Protection Agency. EJScreen: EPA’s Environmental Justice Screening and Mapping Tool (version 2.2). Technical report, U.S. EPA Office of Environmental Justice, 2023. URL <https://www.epa.gov/ejscreen>.
- [22] U.S. Environmental Protection Agency. Reconsideration of the National Ambient Air Quality Standards for Particulate Matter. Technical report, Federal Register 89 FR 16202, 2024.
- [23] World Health Organization. Ambient (outdoor) air pollution – Fact sheet. Technical report, World Health Organization, 2024. URL [https://www.who.int/news-room/fact-sheets/detail/ambient-\(outdoor\)-air-quality-and-health](https://www.who.int/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health).
- [24] Kai Zhang, Jiayi Lin, Yijia Li, Yuxuan Sun, Weihsueh Tong, Fangqun Li, Lung-Chang Chien, Yipei Yang, Wei-Chung Su, Hao Tian, Peng Fu, Fengxiang Qiao, Xiaobo Xue Romeiko, Shao Lin, Siting Luo, and Elena Craft. Unmasking the sky: high-resolution PM2.5 prediction in Texas using machine learning techniques. *Journal of Exposure Science & Environmental Epidemiology*, 34:814–820, 2024.

A Hyperparameters

All models use `random_state=42`. See the public code release for exact configurations. Key hyperparameters are: RF(`n_estimators=200`, `max_depth=12`, `min_samples_leaf=4`); LightGBM(`n_estimators=500`, `lr=0.05`, `max_depth=8`, `num_leaves=31`, `subsample=0.8`, `colsample_bytree=0.8`); XGBoost(`n_estimators=500`, `lr=0.05`, `max_depth=6`, `subsample=0.8`, `colsample_bytree=0.8`); SVR(`kernel=RBF`, `C=1`, `gamma=scale`, `epsilon=0.1`).

B Ablation: Contribution of Autoregressive Features

Ablating the three autoregressive features (`pm25_lag1`, `pm25_lag7`, `pm25_roll7`) while keeping all other features drops pooled LOSO-CV R^2 substantially (by ≥ 0.15 on the ensemble in preliminary experiments) and increases pooled RMSE by over $1 \mu\text{g}/\text{m}^3$. This confirms that in-sensor temporal autocorrelation is a dominant signal for day-of-operation accuracy, and establishes the accuracy ceiling achievable at a cold-start newly-instrumented tract where lag features are unavailable.

C Per-City Ensemble Accuracy

Per-fold LOSO-CV ensemble R^2 is comparable across the four metros, with per-sensor standard deviations dominating any between-city differences. Full per-city breakdown is available in the released code’s `per_sensor_loso` output.

D Training-Set Subsampling for LOSO Throughput

To make the full 4-model 240-fold cross-validation tractable on a single workstation (target wall-clock: under one hour), each fold’s training set is randomly subsampled to 30,000 rows from the approximately 60,000 non-held-out rows, and tree-model `n_estimators` is reduced from 500 to 300 (the values used in the random-split comparison). A preliminary check on a sample of folds confirmed that the resulting per-fold R^2 is within 0.02 of the full-data, full-forest value; the random-split scores in Table 2 use the full training set and full forest size and therefore represent the strongest comparison against LOSO-CV, not a weakened one.

NeurIPS Paper Checklist

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract lists four specific contributions (a 240-fold LOSO-CV benchmark, quantified evaluation bias, fairness-stratified error analysis, and an EJ audit) that are each delivered in a numbered section of the paper. Numerical claims (e.g., $R^2 = 0.810$, $r = 0.875$, 85.1% Q4 exceedance) match the tables and figures.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 6 and Appendix A give exact hyperparameters, random seeds, and training procedure. Section 4 specifies the data sources and preprocessing. Section 5 specifies the LOSO-CV and random-split protocols. We release anonymized data splits and code (Section 9).

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Anonymized data splits, preprocessing pipeline, and training scripts are provided in the supplementary zip. The README specifies exact commands to reproduce each table and figure in the paper. Hyperparameters are enumerated in Appendix A.

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Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 5 specifies both data splits (Random 80/20 with `random_state=42`; LOSO-CV across all 240 sensor folds). Section 6 lists all hyperparameters; we did not conduct hyperparameter search and report default values from prior literature.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: LOSO-CV per-fold standard deviations are reported in Table 2. Figure 6 reports standard errors across sensors within each EJ quartile. The EJ correlation statistics report both r and p -values.

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Justification: PurpleAir data is used under the PurpleAir Terms of Service; EPA EJScreen is public-domain; NOAA GHCN is public-domain; U.S. Census TIGER shapefiles are public-domain. All software dependencies (scikit-learn BSD-3, LightGBM MIT, XGBoost Apache-2.0, matplotlib PSF) are credited in the code release README.

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